

CHAOYANG, China

WMO: 543240

Lat: 41.5478N

Lon: 120.4292E

Elev: 577

StdP: 14.39

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.9	1.9	-21.9	1.7	11.3	-18.2	2.1	12.1	16.7	22.9	15.6	22.8	3.2	180	0.467

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.6	93.5	70.3	90.6	70.4	88.1	69.7	78.1	86.2	76.5	83.8	74.9	81.9	7.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.8	137.9	82.9	74.3	131.1	81.3	72.8	124.3	79.7	42.3	86.3	40.6	84.0	39.0	81.8	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years						
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max				
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
17.7	15.5	13.6	DB	-5.9	98.6	5.1	4.3	-9.6	101.7	-12.6	104.2	-15.5	106.6	-19.2	109.7	
			WB	-6.9	80.0	4.8	1.6	-10.4	81.1	-13.2	82.1	-16.0	82.9	-19.5	84.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.3	17.3	24.9	38.3	54.0	66.1	73.6	78.0	75.6	66.1	52.0	34.7	21.4	(6)
	DBStd	22.53	7.51	8.55	9.18	8.82	7.31	5.59	4.17	4.81	6.13	8.22	9.63	7.50	(7)
	HDD50	3578	1013	702	379	57	1	0	0	0	0	75	465	886	(8)
	HDD65	6577	1478	1122	827	343	74	4	0	1	59	409	909	1351	(9)
	CDD50	3687	0	0	17	178	499	708	868	793	482	136	6	0	(10)
	CDD65	1211	0	0	0	14	107	262	403	329	91	5	0	0	(11)
	CDH74	11437	0	0	4	176	1227	2455	3777	2901	846	51	0	0	(12)
	CDH80	4228	0	0	0	36	439	1004	1484	1060	198	7	0	0	(13)
(14) Wind	WSAvg	5.7	4.7	5.6	7.1	7.8	7.1	6.1	5.5	4.7	4.7	5.1	5.1	4.7	(14)
(15) Precipitation	PrecAvg	18.8	0.1	0.1	0.2	0.8	1.4	3.0	6.4	4.0	1.8	0.7	0.2	0.1	(15)
	PrecMax	27.0	0.4	0.7	1.2	3.3	3.3	5.5	14.8	11.6	5.3	3.0	0.8	0.8	(16)
	PrecMin	10.2	0.0	0.0	0.0	0.0	0.2	0.5	2.1	0.7	0.0	0.0	0.0	0.0	(17)
	PrecStd	4.5	0.1	0.2	0.3	0.7	0.8	1.3	3.6	2.4	1.1	0.7	0.2	0.2	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.1	53.7	72.3	84.7	94.6	98.2	97.6	95.7	88.2	79.3	64.9	46.6	(19)
		MCWB	32.1	39.0	49.3	58.9	63.0	66.4	74.1	72.5	67.0	60.1	48.1	36.7	(20)
	2%	DB	37.5	48.9	64.5	78.6	88.9	92.7	93.0	91.2	84.8	74.3	58.9	41.4	(21)
		MCWB	28.3	35.9	46.0	54.6	61.3	67.3	73.7	73.2	65.6	57.1	45.6	32.6	(22)
	5%	DB	33.9	44.4	58.9	74.5	84.9	89.3	90.1	88.4	82.1	70.3	54.5	37.7	(23)
		MCWB	26.1	33.4	42.7	52.6	59.9	66.3	73.5	72.5	63.9	54.8	43.3	29.7	(24)
	10%	DB	30.4	40.2	54.1	70.0	80.9	86.0	87.6	85.9	79.2	66.5	49.9	34.1	(25)
		MCWB	23.5	30.7	40.2	50.4	58.3	65.7	72.9	72.0	63.0	52.8	40.2	27.2	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.6	40.3	50.8	61.4	68.5	74.8	80.3	79.8	72.9	63.8	51.3	38.4	(27)
		MCDB	42.0	50.6	69.8	78.1	85.8	84.5	89.4	88.7	83.6	73.8	59.3	44.7	(28)
	2%	WB	28.9	36.8	47.1	57.6	65.2	72.0	78.8	77.8	70.1	60.2	46.8	32.9	(29)
		MCDB	37.0	47.1	61.8	73.3	81.9	81.8	87.4	85.5	78.4	69.7	57.1	40.5	(30)
	5%	WB	26.2	33.7	43.7	54.7	63.0	70.5	77.3	76.3	68.0	57.8	43.8	29.9	(31)
		MCDB	33.6	43.9	58.1	70.1	78.1	80.5	85.1	83.3	77.0	67.2	53.4	37.3	(32)
	10%	WB	23.6	30.8	40.5	52.0	61.0	69.0	75.8	74.6	66.0	54.8	40.7	27.2	(33)
		MCDB	30.2	40.0	52.9	66.8	75.1	79.0	82.9	81.5	74.3	64.0	49.3	34.0	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	21.1	21.4	21.7	22.0	22.3	19.4	16.6	17.9	22.4	22.2	19.5	19.5	(35)
		MCDBR	24.9	25.9	28.2	28.2	28.8	25.4	21.4	21.6	26.1	26.7	25.0	23.1	(36)
	5% WB	MCWBR	18.0	17.2	15.8	13.0	10.8	8.4	7.2	7.7	10.0	12.8	14.8	16.3	(37)
		MCWBR	24.3	24.6	26.4	24.2	23.4	18.7	15.9	16.2	20.1	22.6	23.4	22.4	(38)
(40) Clear-Sky Solar Irradiance	taub		0.338	0.405	0.457	0.556	0.609	0.705	0.684	0.595	0.533	0.499	0.400	0.348	(40)
		taud	2.163	1.984	1.898	1.728	1.669	1.554	1.640	1.799	1.863	1.861	2.043	2.150	(41)
	Ebn at Noon		247	244	246	231	222	201	204	219	221	209	219	231	(42)
		Edh at Noon	32	45	56	71	77	87	79	65	57	51	36	30	(43)
(44) All-Sky Solar Radiation	RadAvg		792	1074	1453	1697	1900	1709	1627	1600	1439	1093	788	683	(44)
	RadStd		51	71	92	103	148	147	144	110	79	64	70	42	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	-2.47	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)	N/A	N/A	-1.78	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page